TOOL COST SHEET

EX26-0001 · L mounting bracket

Rev B · sheet metal · 1 cavities · base 3 die set(s) 5040 · try-out on PRESS-100 · tool country IN · costing month 2026-10 · version 2

Uses example rates – not for negotiation

Uses 35 EXAMPLE rates (Rates page) and 51 EXAMPLE master-data rows (Master data). Replace them with real values.

Dies, stage-wise

Die	Operations	Die set · plate material	Die plate L × W × H (mm)	Press (force)	Tool cost (INR)	Tool cost INR	CAPEX INR
Die 1: Blanking	Blanking	4-pillar die set 5040 · C45 / S50C	397 × 150 × 25	PRESS-100 (61.7 t)	188,350	188,350	188,350
Die 2: Piercing	Piercing	4-pillar die set 5040 · C45 / S50C	397 × 150 × 25	PRESS-100 (13.5 t)	178,802	178,802	178,802
Die 3: Bending (V)	Bending (V)	4-pillar die set 5040 · C45 / S50C	397 × 150 × 25	PRESS-100 (0.6 t)	163,759	163,759	163,759

STAGE 1 OF 3

Die 1: Blanking · Blanking

Work area blank / part envelope 326.7 × 80.0 mm · die plate 397 × 150 × 25 mm · die set 4-pillar die set 5040 (C45 / S50C) · press PRESS-100 for 61.7 t · tool cost 188,350 INR = 188,350 INR

Bill of material

Element	Category	Qty	Grade	Make	Standard part / part no.	Weight each (kg)	Line total
Die set 4-pillar die set 5040 •	bought out	1	C45 / S50C	Local die-set maker / Misumi		260.00	68,000
Die plate (cutting) •	made	1	D2 (1.2379)	Bohler / Daido		11.45	5,727
Blanking / trimming punch •	made	1	D2 (1.2379)	Bohler / Daido		6.04	3,019
Punch holder plate •	made	1	C45 / S50C	JSW / Tata		10.28	1,130
Stripper plate •	made	1	C45 / S50C	JSW / Tata		9.34	1,028
Backing plate (hardened) •	made	2	Mould base steel (4140 / 1.7225)	LKM / Futaba		5.61	1,570
Stock guides & stopper •	made	1	C45 / S50C	JSW / Tata		2.80	308
Die springs / urethane •	bought out	8		Danly / Misumi			3,600
Screws, dowels & bolts (set) •	bought out	1		Unbrako			2,095
Wire cut - die profile •	process	1					3,942
Wire cut - punches •	process	1					8,315
CNC machining - plates & forming steels •	process	1					5,787
Surface grinding •	process	1					5,627
Heat treatment (die steels) •	process	1					3,485
Fitting & assembly •	process	1					13,624
Die design •	process	1					8,524

STAGE 2 OF 3

Die 2: Piercing · Piercing

Work area blank / part envelope 326.7 × 80.0 mm · die plate 397 × 150 × 25 mm · die set 4-pillar die set 5040 (C45 / S50C) · press PRESS-100 for 13.5 t · tool cost 178,802 INR = 178,802 INR

Bill of material

Element	Category	Qty	Grade	Make	Standard part / part no.	Weight each (kg)	Line total
Die set 4-pillar die set 5040 •	bought out	1	C45 / S50C	Local die-set maker / Misumi		260.00	68,000
Die plate (cutting) •	made	1	D2 (1.2379)	Bohler / Daido		11.45	5,727
Piercing punch (standard) •	bought out	6		Misumi / Punch			2,580
Punch holder plate •	made	1	C45 / S50C	JSW / Tata		10.28	1,130
Stripper plate •	made	1	C45 / S50C	JSW / Tata		9.34	1,028
Backing plate (hardened) •	made	2	Mould base steel (4140 / 1.7225)	LKM / Futaba		5.61	1,570
Stock guides & stopper •	made	1	C45 / S50C	JSW / Tata		2.80	308
Die springs / urethane •	bought out	8		Danly / Misumi			3,600
Screws, dowels & bolts (set) •	bought out	1		Unbrako			2,095
Wire cut - die profile •	process	1					1,500
Wire cut - punches •	process	1					2,456
CNC machining - plates & forming steels •	process	1					5,787
Surface grinding •	process	1					5,627
Heat treatment (die steels) •	process	1					2,747
Fitting & assembly •	process	1					15,688
Die design •	process	1					8,524

STAGE 3 OF 3

Die 3: Bending (V) · Bending (V)

Work area blank / part envelope 326.7 × 80.0 mm · die plate 397 × 150 × 25 mm · die set 4-pillar die set 5040 (C45 / S50C) · press PRESS-100 for 0.6 t · tool cost 163,759 INR = 163,759 INR

Bill of material

Element	Category	Qty	Grade	Make	Standard part / part no.	Weight each (kg)	Line total
Die set 4-pillar die set 5040 •	bought out	1	C45 / S50C	Local die-set maker / Misumi		260.00	68,000
Punch holder plate •	made	1	C45 / S50C	JSW / Tata		10.28	1,130
Backing plate (hardened) •	made	2	Mould base steel (4140 / 1.7225)	LKM / Futaba		5.61	1,570
Bending punch •	made	1	D2 (1.2379)	Bohler / Daido		0.99	493
Bending die (V-block) •	made	1	D2 (1.2379)	Bohler / Daido		5.08	2,541
Pressure pad •	made	1	C45 / S50C	JSW / Tata		9.34	1,028
Stock guides & stopper •	made	1	C45 / S50C	JSW / Tata		2.80	308
Die springs / urethane •	bought out	8		Danly / Misumi			3,600
Screws, dowels & bolts (set) •	bought out	1		Unbrako			2,095
CNC machining - plates & forming steels •	process	1					5,787
Surface grinding •	process	1					5,627
Heat treatment (die steels) •	process	1					2,747
Fitting & assembly •	process	1					13,211
Die design •	process	1					8,524

Result

	INR	INR
Tool cost per tool	**530,911**	**530,911**
Tool CAPEX (1 tools over the programme)	530,911	530,911

Cost build-up (process-wise, then trial, overhead, profit, packing, transport, warranty)

Step	Formula	Value	Unit
Machining (CNC, EDM, wire cut, grinding, drilling) (all dies)	Σ BOM lines of the group	50,453	INR
Heat & surface treatment (all dies)	Σ BOM lines of the group	8,978	INR
Fitting & assembly (all dies)	Σ BOM lines of the group	42,523	INR
Design (all dies)	Σ BOM lines of the group	25,571	INR
Die set: tool cost of all dies	d0 + d1 + d2	**530,911**	**INR**
Tool CAPEX: all dies × tools over the programme	d0 + d1 + d2	530,911	INR

BOM & processes (INR; • = EXAMPLE master-data row)

Element	Category	Qty	Steel	Steel cost	Bought-out	Process	Hours / amount	Process cost	Line total
Die set 4-pillar die set 5040 •	bought out	1	C45 / S50C		68,000				68,000
Die plate (cutting) •	made	1	D2 (1.2379)	5,727					5,727
Blanking / trimming punch •	made	1	D2 (1.2379)	3,019					3,019
Punch holder plate •	made	1	C45 / S50C	1,130					1,130
Stripper plate •	made	1	C45 / S50C	1,028					1,028
Backing plate (hardened) •	made	2	Mould base steel (4140 / 1.7225)	1,570					1,570
Stock guides & stopper •	made	1	C45 / S50C	308.2954					308.2954
Die springs / urethane •	bought out	8			3,600				3,600
Screws, dowels & bolts (set) •	bought out	1			2,095				2,095
Wire cut - die profile •	process	1				Wire cut	4.8198	3,942	3,942
Wire cut - punches •	process	1				Wire cut	10.1675	8,315	8,315

Element	Category	Qty	Steel	Steel cost	Bought-out	Process	Hours / amount	Process cost	Line total
CNC machining - plates & forming steels •	process	1				CNC milling	7.9505	5,787	5,787
Surface grinding •	process	1				Grinding	12.901	5,627	5,627
Heat treatment (die steels) •	process	1				Heat treatment (per kg)	29.0436	3,485	3,485
Fitting & assembly •	process	1				Fitting & assembly	16.5	13,624	13,624
Die design •	process	1				Design	18	8,524	8,524
Die set 4-pillar die set 5040 •	bought out	1	C45 / S50C		68,000				68,000
Die plate (cutting) •	made	1	D2 (1.2379)	5,727					5,727
Piercing punch (standard) •	bought out	6			2,580				2,580
Punch holder plate •	made	1	C45 / S50C	1,130					1,130
Stripper plate •	made	1	C45 / S50C	1,028					1,028
Backing plate (hardened) •	made	2	Mould base steel (4140 / 1.7225)	1,570					1,570
Stock guides & stopper •	made	1	C45 / S50C	308.2954					308.2954
Die springs / urethane •	bought out	8			3,600				3,600
Screws, dowels & bolts (set) •	bought out	1			2,095				2,095
Wire cut - die profile •	process	1				Wire cut	1.8345	1,500	1,500
Wire cut - punches •	process	1				Wire cut	3.0028	2,456	2,456
CNC machining - plates & forming steels •	process	1				CNC milling	7.9505	5,787	5,787
Surface grinding •	process	1				Grinding	12.901	5,627	5,627
Heat treatment (die steels) •	process	1				Heat treatment (per kg)	22.8886	2,747	2,747
Fitting & assembly •	process	1				Fitting & assembly	19	15,688	15,688
Die design •	process	1				Design	18	8,524	8,524
Die set 4-pillar die set 5040 •	bought out	1	C45 / S50C		68,000				68,000
Punch holder plate •	made	1	C45 / S50C	1,130					1,130
Backing plate (hardened) •	made	2	Mould base steel (4140 / 1.7225)	1,570					1,570
Bending punch •	made	1	D2 (1.2379)	492.8					492.8
Bending die (V-block) •	made	1	D2 (1.2379)	2,541					2,541

Element	Category	Qty	Steel	Steel cost	Bought-out	Process	Hours / amount	Process cost	Line total
Pressure pad •	made	1	C45 / S50C	1,028					1,028
Stock guides & stopper •	made	1	C45 / S50C	308.2954					308.2954
Die springs / urethane •	bought out	8			3,600				3,600
Screws, dowels & bolts (set) •	bought out	1			2,095				2,095
CNC machining - plates & forming steels •	process	1				CNC milling	7.9505	5,787	5,787
Surface grinding •	process	1				Grinding	12.901	5,627	5,627
Heat treatment (die steels) •	process	1				Heat treatment (per kg)	22.8886	2,747	2,747
Fitting & assembly •	process	1				Fitting & assembly	16	13,211	13,211
Die design •	process	1				Design	18	8,524	8,524

Every step and formula

Step	Formula	Value	Unit
Die set 4-pillar die set 5040 · plate steel C45 / S50C (value)	weight * rate	28,600	INR
Die plate (cutting) · quantity	1	1	pcs
Shots (strokes) over the programme per tool	annual_volume * programme_years / cavities	1,000,000	shots
Die plate height	max(die_plate_min_mm, die_plate_t_factor * t)	25	mm
Die plate length	work_l + 2 * die_wall_mm	396.7	mm
Die plate width	work_w + 2 * die_wall_mm	150	mm
Die plate (cutting) · weight each	die_l * die_w * die_h * density / 1e6	11.4547	kg
Die plate (cutting) · steel weight	qty * weight	11.4547	kg
Die plate (cutting) · steel D2 (1.2379)	qty * weight * rate	5,727	INR
Die plate (cutting)	x0	5,727	INR
Blanking / trimming punch · quantity	op_blank + op_trim	1	pcs
Blanking / trimming punch · weight each	blank_l * blank_w * punch_len_mm * density / 1e6	6.0374	kg
Blanking / trimming punch · steel weight	qty * weight	6.0374	kg
Blanking / trimming punch · steel D2 (1.2379)	qty * weight * rate	3,019	INR
Blanking / trimming punch	x0	3,019	INR

Step	Formula	Value	Unit
Punch holder plate · quantity	1	1	pcs
Punch holder plate · weight each	die_l * die_w * punch_plate_h_mm * density / 1e6	10.2765	kg
Punch holder plate · steel weight	qty * weight	10.2765	kg
Punch holder plate · steel C45 / S50C	qty * weight * rate	1,130	INR
Punch holder plate	x0	1,130	INR
Stripper plate · quantity	1	1	pcs
Stripper plate · weight each	die_l * die_w * stripper_h_mm * density / 1e6	9.3423	kg
Stripper plate · steel weight	qty * weight	9.3423	kg
Stripper plate · steel C45 / S50C	qty * weight * rate	1,028	INR
Stripper plate	x0	1,028	INR
Backing plate (hardened) · quantity	2	2	pcs
Backing plate (hardened) · weight each	die_l * die_w * backing_h_mm * density / 1e6	5.6054	kg
Backing plate (hardened) · steel weight	qty * weight	11.2107	kg
Backing plate (hardened) · steel Mould base steel (4140 / 1.7225)	qty * weight * rate	1,570	INR
Backing plate (hardened)	x0	1,570	INR
Stock guides & stopper · quantity	1	1	pcs
Stock guides & stopper · weight each	2 * die_l * 30 * 15 * density / 1e6	2.8027	kg
Stock guides & stopper · steel weight	qty * weight	2.8027	kg
Stock guides & stopper · steel C45 / S50C	qty * weight * rate	308.2954	INR
Stock guides & stopper	x0	308.2954	INR
Die springs / urethane · quantity	4 + 2 * ceil(die_l / 200)	8	pcs
Die springs / urethane · catalogue price each	450	450	INR
Die springs / urethane · bought-out	qty * each	3,600	INR
Die springs / urethane	x0	3,600	INR
Screws, dowels & bolts (set) · quantity	1	1	pcs
Screws, dowels & bolts (set) · catalogue price each	1500 + die_l * die_w / 100	2,095	INR
Screws, dowels & bolts (set) · bought-out	qty * each	2,095	INR

Step	Formula	Value	Unit
Screws, dowels & bolts (set)	x0	2,095	INR
Wire cut - die profile · quantity	1	1	pcs
Cutting length in this die	op_blank * blank_perimeter + op_pierce * hole_perimeter + op_trim * blank_perimeter	733.4	mm
Wire cut - die profile · hours each	wirecut_setup_h + cut_len * die_h / wirecut_mm2_per_h	4.8198	h
Wire cut - die profile · hours	qty * hours	4.8198	h
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	600,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	540,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	140.625	INR/h
Average investment	(machine_price + residual_value) / 2	3,300,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	85.9375	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	59.5	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	78.125	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	8.75	INR/h
Consumables & tooling	consumables_rate	250	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	622.9375	INR/h
Overhead	machine_cost * overhead_pct / 100	93.4406	INR/h
MHR WEDM · Wire EDM	machine_cost + overhead	716.3781	INR/h
Machine hour rate (MHR) in INR	value * fx	716.3781	INR/h
Paid hours per year	working_days * hours_per_shift	2,400	h/yr
Annual salary / CTC	monthly_salary * 12	360,000	INR/yr
Employer extra cost	annual_salary * employer_extra_pct / 100	0	INR/yr
Base rate	(annual_salary + employer_cost) / paid_hours_per_year	150	INR/h
Supervision & labour overhead	base_rate * supervision_pct / 100	22.5	INR/h
Non-productive time	(base_rate + supervision) * (100 / productive_pct - 1)	30.4412	INR/h
Labour rate · skilled_machinist	base_rate + supervision + productivity_loss	202.9412	INR/h
Labour rate in INR	value * fx	202.9412	INR/h

Step	Formula	Value	Unit
Wire cut: cost per hour	mhr + operators * labour	817.8487	INR/h
Wire cut - die profile · Wire cut	hours * cost_per_hour	3,942	INR
Wire cut - die profile	x0	3,942	INR
Wire cut - punches · quantity	1	1	pcs
Wire cut - punches · hours each	wirecut_setup_h + cut_len * punch_len_mm / wirecut_mm2_per_h	10.1675	h
Wire cut - punches · hours	qty * hours	10.1675	h
Wire cut - punches · Wire cut	hours * cost_per_hour	8,315	INR
Wire cut - punches	x0	8,315	INR
CNC machining - plates & forming steels · quantity	1	1	pcs
CNC machining - plates & forming steels · hours each	cnc_setup_h + 4 * die_l * die_w / 100 / cnc_plate_cm2_per_h	7.9505	h
CNC machining - plates & forming steels · hours	qty * hours	7.9505	h
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	450,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	405,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	105.4688	INR/h
Average investment	(machine_price + residual_value) / 2	2,475,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	64.4531	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	76.5	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	46.875	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	13.125	INR/h
Consumables & tooling	consumables_rate	150	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	456.4219	INR/h
Overhead	machine_cost * overhead_pct / 100	68.4633	INR/h
MHR VMC-850 · CNC vertical machining centre 850	machine_cost + overhead	524.8852	INR/h
Machine hour rate (MHR) in INR	value * fx	524.8852	INR/h
CNC milling: cost per hour	mhr + operators * labour	727.8263	INR/h
CNC machining - plates & forming steels · CNC milling	hours * cost_per_hour	5,787	INR

Step	Formula	Value	Unit
CNC machining - plates & forming steels	x0	5,787	INR
Surface grinding · quantity	1	1	pcs
Surface grinding · hours each	grind_setup_h + 6 * die_l * die_w / 100 / grind_cm2_per_h	12.901	h
Surface grinding · hours	qty * hours	12.901	h
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	90,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	142,500	INR/yr
Depreciation	depreciation_per_year / annual_hours	37.1094	INR/h
Average investment	(machine_price + residual_value) / 2	945,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	24.6094	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	38.25	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	14.0625	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	8.75	INR/h
Consumables & tooling	consumables_rate	80	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	202.7812	INR/h
Overhead	machine_cost * overhead_pct / 100	30.4172	INR/h
MHR GRIND-SURF · Surface grinder	machine_cost + overhead	233.1984	INR/h
Machine hour rate (MHR) in INR	value * fx	233.1984	INR/h
Grinding: cost per hour	mhr + operators * labour	436.1396	INR/h
Surface grinding · Grinding	hours * cost_per_hour	5,627	INR
Surface grinding	x0	5,627	INR
Heat treatment (die steels) · quantity	1	1	pcs
Heat treatment (die steels) · amount each	die_l * die_w * (die_h + 2 * backing_h_mm) * 7.85 / 1e6 + (op_blank + op_trim) * blank_l * blank_w * punch_len_mm * 7.85 / 1e6	29.0436	kg
Heat treatment (die steels) · Heat treatment (per kg)	max(min_charge, qty * amount * rate)	3,485	INR
Heat treatment (die steels)	x0	3,485	INR
Fitting & assembly · quantity	1	1	pcs
Cutting punches in this die	op_blank * 1 + op_pierce * hole_count + op_trim * 1	1	pcs

Step	Formula	Value	Unit
Fitting & assembly · hours each	assy_h_base + assy_h_per_punch * punches	16.5	h
Fitting & assembly · hours	qty * hours	16.5	h
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	350,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	210,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	54.6875	INR/h
Average investment	(machine_price + residual_value) / 2	1,925,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	50.1302	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	19.125	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	18.2292	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	16.4062	INR/h
Consumables & tooling	consumables_rate	30	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	188.5781	INR/h
Overhead	machine_cost * overhead_pct / 100	28.2867	INR/h
MHR SPOTTING · Spotting / try-out press	machine_cost + overhead	216.8648	INR/h
Machine hour rate (MHR) in INR	value * fx	216.8648	INR/h
Paid hours per year	working_days * hours_per_shift	2,400	h/yr
Annual salary / CTC	monthly_salary * 12	540,000	INR/yr
Employer extra cost	annual_salary * employer_extra_pct / 100	0	INR/yr
Base rate	(annual_salary + employer_cost) / paid_hours_per_year	225	INR/h
Supervision & labour overhead	base_rate * supervision_pct / 100	33.75	INR/h
Non-productive time	(base_rate + supervision) * (100 / productive_pct - 1)	45.6618	INR/h
Labour rate · toolmaker	base_rate + supervision + productivity_loss	304.4118	INR/h
Labour rate in INR	value * fx	304.4118	INR/h
Fitting & assembly: cost per hour	mhr + operators * labour	825.6884	INR/h
Fitting & assembly · Fitting & assembly	hours * cost_per_hour	13,624	INR
Fitting & assembly	x0	13,624	INR

Step	Formula	Value	Unit
Die design · quantity	1	1	pcs
Die design · hours each	design_h_base + design_h_per_op * die_ops	18	h
Die design · hours	qty * hours	18	h
Paid hours per year	working_days * hours_per_shift	2,400	h/yr
Annual salary / CTC	monthly_salary * 12	840,000	INR/yr
Employer extra cost	annual_salary * employer_extra_pct / 100	0	INR/yr
Base rate	(annual_salary + employer_cost) / paid_hours_per_year	350	INR/h
Supervision & labour overhead	base_rate * supervision_pct / 100	52.5	INR/h
Non-productive time	(base_rate + supervision) * (100 / productive_pct - 1)	71.0294	INR/h
Labour rate · designer	base_rate + supervision + productivity_loss	473.5294	INR/h
Labour rate in INR	value * fx	473.5294	INR/h
Design: cost per hour	operators * labour	473.5294	INR/h
Die design · Design	hours * cost_per_hour	8,524	INR
Die design	x0	8,524	INR
Machining (CNC, EDM, wire cut, grinding, drilling)	x0 + x1 + x2 + x3	23,671	INR
Heat & surface treatment	x0	3,485	INR
Fitting & assembly	x0	13,624	INR
Design	x0	8,524	INR
Tool steel	x0 + x1 + x2 + x3 + x4 + x5	12,782	INR
Mould base & standard parts	x0 + x1 + x2	73,695	INR
Processes (design excluded)	x0 + x1 + x2	40,780	INR
Design	x0	8,524	INR
Tool weight	x0 + x1 + x2 + x3 + x4 + x5 + x6	311.1244	kg
Shot weight (net)	cavities * part_weight_g / 1000	0.1998	kg
Press force needed for this die	die_force_t * safety	61.6984	t
Try-out machine: PRESS-100 · Power press 100 t	smallest 'Press' machine in Master data → Machines whose tonnage ≥ the press force	100	t
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr

Step	Formula	Value	Unit
Residual value	machine_price * residual_pct / 100	300,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	180,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	46.875	INR/h
Average investment	(machine_price + residual_value) / 2	1,650,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	42.9688	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	56.1	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	31.25	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	8.75	INR/h
Consumables & tooling	consumables_rate	60	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	245.9437	INR/h
Overhead	machine_cost * overhead_pct / 100	36.8916	INR/h
MHR PRESS-100 · Power press 100 t	machine_cost + overhead	282.8353	INR/h
Machine hour rate (MHR) in INR	value * fx	282.8353	INR/h
Try-out crew per hour (operators × labour rate)	operators * labour	304.4118	INR/h
Try-out hours	trials * trial_hours_each	6	h
Try-out machine & crew	trial_hours * (trial_mhr + trial_labour_per_h)	3,523	INR
Try-out material	trials * trial_shots_each * shot_weight_kg * trial_material_rate	5,755	INR
Try-out	trial_machine_cost + trial_material_cost	9,279	INR
Manufacturing cost	material_cost + bought_out_cost + process_cost + design_cost + trial_cost	145,059	INR
Toolmaker overhead	manufacturing_cost * overhead_pct / 100	17,407	INR
Toolmaker profit	(manufacturing_cost + tool_overhead) * profit_pct / 100	16,247	INR
Tool price ex-works	manufacturing_cost + tool_overhead + tool_profit	178,712	INR
Packing	tool_price * packing_pct / 100	1,787	INR
Transport	tool_weight_kg * transport_rate	2,489	INR
Warranty / spares	tool_price * warranty_pct / 100 * warranty_on	5,361	INR
Die 1: Blanking · tool cost	tool_price + tool_packing + tool_transport + tool_warranty	188,350	INR
Die 1: Blanking · tool CAPEX	tool_cost * tools_count	188,350	INR

Step	Formula	Value	Unit
Tool cost per tool in INR	value * fx	188,350	INR
Tool CAPEX in INR	value * fx	188,350	INR
Die set 4-pillar die set 5040 · plate steel C45 / S50C (value)	weight * rate	28,600	INR
Die plate (cutting) · quantity	1	1	pcs
Shots (strokes) over the programme per tool	annual_volume * programme_years / cavities	1,000,000	shots
Die plate height	max(die_plate_min_mm, die_plate_t_factor * t)	25	mm
Die plate length	work_l + 2 * die_wall_mm	396.7	mm
Die plate width	work_w + 2 * die_wall_mm	150	mm
Die plate (cutting) · weight each	die_l * die_w * die_h * density / 1e6	11.4547	kg
Die plate (cutting) · steel weight	qty * weight	11.4547	kg
Die plate (cutting) · steel D2 (1.2379)	qty * weight * rate	5,727	INR
Die plate (cutting)	x0	5,727	INR
Piercing punch (standard) · quantity	op_pierce * hole_count	6	pcs
Piercing punch (standard) · catalogue price each	350 + 40 * t	430	INR
Piercing punch (standard) · bought-out	qty * each	2,580	INR
Piercing punch (standard)	x0	2,580	INR
Punch holder plate · quantity	1	1	pcs
Punch holder plate · weight each	die_l * die_w * punch_plate_h_mm * density / 1e6	10.2765	kg
Punch holder plate · steel weight	qty * weight	10.2765	kg
Punch holder plate · steel C45 / S50C	qty * weight * rate	1,130	INR
Punch holder plate	x0	1,130	INR
Stripper plate · quantity	1	1	pcs
Stripper plate · weight each	die_l * die_w * stripper_h_mm * density / 1e6	9.3423	kg
Stripper plate · steel weight	qty * weight	9.3423	kg
Stripper plate · steel C45 / S50C	qty * weight * rate	1,028	INR
Stripper plate	x0	1,028	INR
Backing plate (hardened) · quantity	2	2	pcs

Step	Formula	Value	Unit
Backing plate (hardened) · weight each	die_l * die_w * backing_h_mm * density / 1e6	5.6054	kg
Backing plate (hardened) · steel weight	qty * weight	11.2107	kg
Backing plate (hardened) · steel Mould base steel (4140 / 1.7225)	qty * weight * rate	1,570	INR
Backing plate (hardened)	x0	1,570	INR
Stock guides & stopper · quantity	1	1	pcs
Stock guides & stopper · weight each	2 * die_l * 30 * 15 * density / 1e6	2.8027	kg
Stock guides & stopper · steel weight	qty * weight	2.8027	kg
Stock guides & stopper · steel C45 / S50C	qty * weight * rate	308.2954	INR
Stock guides & stopper	x0	308.2954	INR
Die springs / urethane · quantity	4 + 2 * ceil(die_l / 200)	8	pcs
Die springs / urethane · catalogue price each	450	450	INR
Die springs / urethane · bought-out	qty * each	3,600	INR
Die springs / urethane	x0	3,600	INR
Screws, dowels & bolts (set) · quantity	1	1	pcs
Screws, dowels & bolts (set) · catalogue price each	1500 + die_l * die_w / 100	2,095	INR
Screws, dowels & bolts (set) · bought-out	qty * each	2,095	INR
Screws, dowels & bolts (set)	x0	2,095	INR
Wire cut - die profile · quantity	1	1	pcs
Cutting length in this die	op_blank * blank_perimeter + op_pierce * hole_perimeter + op_trim * blank_perimeter	160.2212	mm
Wire cut - die profile · hours each	wirecut_setup_h + cut_len * die_h / wirecut_mm2_per_h	1.8345	h
Wire cut - die profile · hours	qty * hours	1.8345	h
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	600,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	540,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	140.625	INR/h
Average investment	(machine_price + residual_value) / 2	3,300,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	85.9375	INR/h

Step	Formula	Value	Unit
Power	power_kw * load_factor_pct / 100 * electricity_rate	59.5	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	78.125	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	8.75	INR/h
Consumables & tooling	consumables_rate	250	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	622.9375	INR/h
Overhead	machine_cost * overhead_pct / 100	93.4406	INR/h
MHR WEDM · Wire EDM	machine_cost + overhead	716.3781	INR/h
Machine hour rate (MHR) in INR	value * fx	716.3781	INR/h
Paid hours per year	working_days * hours_per_shift	2,400	h/yr
Annual salary / CTC	monthly_salary * 12	360,000	INR/yr
Employer extra cost	annual_salary * employer_extra_pct / 100	0	INR/yr
Base rate	(annual_salary + employer_cost) / paid_hours_per_year	150	INR/h
Supervision & labour overhead	base_rate * supervision_pct / 100	22.5	INR/h
Non-productive time	(base_rate + supervision) * (100 / productive_pct - 1)	30.4412	INR/h
Labour rate · skilled_machinist	base_rate + supervision + productivity_loss	202.9412	INR/h
Labour rate in INR	value * fx	202.9412	INR/h
Wire cut: cost per hour	mhr + operators * labour	817.8487	INR/h
Wire cut - die profile · Wire cut	hours * cost_per_hour	1,500	INR
Wire cut - die profile	x0	1,500	INR
Wire cut - punches · quantity	1	1	pcs
Wire cut - punches · hours each	wirecut_setup_h + cut_len * punch_len_mm / wirecut_mm2_per_h	3.0028	h
Wire cut - punches · hours	qty * hours	3.0028	h
Wire cut - punches · Wire cut	hours * cost_per_hour	2,456	INR
Wire cut - punches	x0	2,456	INR
CNC machining - plates & forming steels · quantity	1	1	pcs
CNC machining - plates & forming steels · hours each	cnc_setup_h + 4 * die_l * die_w / 100 / cnc_plate_cm2_per_h	7.9505	h
CNC machining - plates & forming steels · hours	qty * hours	7.9505	h

Step	Formula	Value	Unit
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	450,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	405,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	105.4688	INR/h
Average investment	(machine_price + residual_value) / 2	2,475,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	64.4531	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	76.5	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	46.875	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	13.125	INR/h
Consumables & tooling	consumables_rate	150	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	456.4219	INR/h
Overhead	machine_cost * overhead_pct / 100	68.4633	INR/h
MHR VMC-850 · CNC vertical machining centre 850	machine_cost + overhead	524.8852	INR/h
Machine hour rate (MHR) in INR	value * fx	524.8852	INR/h
CNC milling: cost per hour	mhr + operators * labour	727.8263	INR/h
CNC machining - plates & forming steels · CNC milling	hours * cost_per_hour	5,787	INR
CNC machining - plates & forming steels	x0	5,787	INR
Surface grinding · quantity	1	1	pcs
Surface grinding · hours each	grind_setup_h + 6 * die_l * die_w / 100 / grind_cm2_per_h	12.901	h
Surface grinding · hours	qty * hours	12.901	h
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	90,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	142,500	INR/yr
Depreciation	depreciation_per_year / annual_hours	37.1094	INR/h
Average investment	(machine_price + residual_value) / 2	945,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	24.6094	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	38.25	INR/h

Step	Formula	Value	Unit
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	14.0625	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	8.75	INR/h
Consumables & tooling	consumables_rate	80	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	202.7812	INR/h
Overhead	machine_cost * overhead_pct / 100	30.4172	INR/h
MHR GRIND-SURF · Surface grinder	machine_cost + overhead	233.1984	INR/h
Machine hour rate (MHR) in INR	value * fx	233.1984	INR/h
Grinding: cost per hour	mhr + operators * labour	436.1396	INR/h
Surface grinding · Grinding	hours * cost_per_hour	5,627	INR
Surface grinding	x0	5,627	INR
Heat treatment (die steels) · quantity	1	1	pcs
Heat treatment (die steels) · amount each	die_l * die_w * (die_h + 2 * backing_h_mm) * 7.85 / 1e6 + (op_blank + op_trim) * blank_l * blank_w * punch_len_mm * 7.85 / 1e6	22.8886	kg
Heat treatment (die steels) · Heat treatment (per kg)	max(min_charge, qty * amount * rate)	2,747	INR
Heat treatment (die steels)	x0	2,747	INR
Fitting & assembly · quantity	1	1	pcs
Cutting punches in this die	op_blank * 1 + op_pierce * hole_count + op_trim * 1	6	pcs
Fitting & assembly · hours each	assy_h_base + assy_h_per_punch * punches	19	h
Fitting & assembly · hours	qty * hours	19	h
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	350,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	210,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	54.6875	INR/h
Average investment	(machine_price + residual_value) / 2	1,925,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	50.1302	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	19.125	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	18.2292	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	16.4062	INR/h

Step	Formula	Value	Unit
Consumables & tooling	consumables_rate	30	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	188.5781	INR/h
Overhead	machine_cost * overhead_pct / 100	28.2867	INR/h
MHR SPOTTING · Spotting / try-out press	machine_cost + overhead	216.8648	INR/h
Machine hour rate (MHR) in INR	value * fx	216.8648	INR/h
Paid hours per year	working_days * hours_per_shift	2,400	h/yr
Annual salary / CTC	monthly_salary * 12	540,000	INR/yr
Employer extra cost	annual_salary * employer_extra_pct / 100	0	INR/yr
Base rate	(annual_salary + employer_cost) / paid_hours_per_year	225	INR/h
Supervision & labour overhead	base_rate * supervision_pct / 100	33.75	INR/h
Non-productive time	(base_rate + supervision) * (100 / productive_pct - 1)	45.6618	INR/h
Labour rate · toolmaker	base_rate + supervision + productivity_loss	304.4118	INR/h
Labour rate in INR	value * fx	304.4118	INR/h
Fitting & assembly: cost per hour	mhr + operators * labour	825.6884	INR/h
Fitting & assembly · Fitting & assembly	hours * cost_per_hour	15,688	INR
Fitting & assembly	x0	15,688	INR
Die design · quantity	1	1	pcs
Die design · hours each	design_h_base + design_h_per_op * die_ops	18	h
Die design · hours	qty * hours	18	h
Paid hours per year	working_days * hours_per_shift	2,400	h/yr
Annual salary / CTC	monthly_salary * 12	840,000	INR/yr
Employer extra cost	annual_salary * employer_extra_pct / 100	0	INR/yr
Base rate	(annual_salary + employer_cost) / paid_hours_per_year	350	INR/h
Supervision & labour overhead	base_rate * supervision_pct / 100	52.5	INR/h
Non-productive time	(base_rate + supervision) * (100 / productive_pct - 1)	71.0294	INR/h
Labour rate · designer	base_rate + supervision + productivity_loss	473.5294	INR/h
Labour rate in INR	value * fx	473.5294	INR/h

Step	Formula	Value	Unit
Design: cost per hour	operators * labour	473.5294	INR/h
Die design · Design	hours * cost_per_hour	8,524	INR
Die design	x0	8,524	INR
Machining (CNC, EDM, wire cut, grinding, drilling)	x0 + x1 + x2 + x3	15,369	INR
Heat & surface treatment	x0	2,747	INR
Fitting & assembly	x0	15,688	INR
Design	x0	8,524	INR
Tool steel	x0 + x1 + x2 + x3 + x4	9,763	INR
Mould base & standard parts	x0 + x1 + x2 + x3	76,275	INR
Processes (design excluded)	x0 + x1 + x2	33,804	INR
Design	x0	8,524	INR
Tool weight	x0 + x1 + x2 + x3 + x4 + x5	305.0869	kg
Shot weight (net)	cavities * part_weight_g / 1000	0.1998	kg
Press force needed for this die	die_force_t * safety	13.4789	t
Try-out machine: PRESS-100 · Power press 100 t	smallest 'Press' machine in Master data → Machines whose tonnage ≥ the press force	100	t
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	300,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	180,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	46.875	INR/h
Average investment	(machine_price + residual_value) / 2	1,650,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	42.9688	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	56.1	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	31.25	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	8.75	INR/h
Consumables & tooling	consumables_rate	60	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	245.9437	INR/h
Overhead	machine_cost * overhead_pct / 100	36.8916	INR/h

Step	Formula	Value	Unit
MHR PRESS-100 · Power press 100 t	machine_cost + overhead	282.8353	INR/h
Machine hour rate (MHR) in INR	value * fx	282.8353	INR/h
Try-out crew per hour (operators × labour rate)	operators * labour	304.4118	INR/h
Try-out hours	trials * trial_hours_each	6	h
Try-out machine & crew	trial_hours * (trial_mhr + trial_labour_per_h)	3,523	INR
Try-out material	trials * trial_shots_each * shot_weight_kg * trial_material_rate	5,755	INR
Try-out	trial_machine_cost + trial_material_cost	9,279	INR
Manufacturing cost	material_cost + bought_out_cost + process_cost + design_cost + trial_cost	137,644	INR
Toolmaker overhead	manufacturing_cost * overhead_pct / 100	16,517	INR
Toolmaker profit	(manufacturing_cost + tool_overhead) * profit_pct / 100	15,416	INR
Tool price ex-works	manufacturing_cost + tool_overhead + tool_profit	169,578	INR
Packing	tool_price * packing_pct / 100	1,696	INR
Transport	tool_weight_kg * transport_rate	2,441	INR
Warranty / spares	tool_price * warranty_pct / 100 * warranty_on	5,087	INR
Die 2: Piercing · tool cost	tool_price + tool_packing + tool_transport + tool_warranty	178,802	INR
Die 2: Piercing · tool CAPEX	tool_cost * tools_count	178,802	INR
Tool cost per tool in INR	value * fx	178,802	INR
Tool CAPEX in INR	value * fx	178,802	INR
Die set 4-pillar die set 5040 · plate steel C45 / S50C (value)	weight * rate	28,600	INR
Punch holder plate · quantity	1	1	pcs
Die plate length	work_l + 2 * die_wall_mm	396.7	mm
Die plate width	work_w + 2 * die_wall_mm	150	mm
Punch holder plate · weight each	die_l * die_w * punch_plate_h_mm * density / 1e6	10.2765	kg
Punch holder plate · steel weight	qty * weight	10.2765	kg
Punch holder plate · steel C45 / S50C	qty * weight * rate	1,130	INR
Punch holder plate	x0	1,130	INR
Backing plate (hardened) · quantity	2	2	pcs

Step	Formula	Value	Unit
Backing plate (hardened) · weight each	die_l * die_w * backing_h_mm * density / 1e6	5.6054	kg
Backing plate (hardened) · steel weight	qty * weight	11.2107	kg
Backing plate (hardened) · steel Mould base steel (4140 / 1.7225)	qty * weight * rate	1,570	INR
Backing plate (hardened)	x0	1,570	INR
Bending punch · quantity	op_bend	1	pcs
Bending punch · weight each	blank_w * 50 * (60 + 2 * t) * density / 1e6	0.9856	kg
Bending punch · steel weight	qty * weight	0.9856	kg
Bending punch · steel D2 (1.2379)	qty * weight * rate	492.8	INR
Bending punch	x0	492.8	INR
Bending die (V-block) · quantity	op_bend	1	pcs
Bending die (V-block) · weight each	(blank_w + 2 * die_wall_mm) * 100 * 60 * density / 1e6	5.082	kg
Bending die (V-block) · steel weight	qty * weight	5.082	kg
Bending die (V-block) · steel D2 (1.2379)	qty * weight * rate	2,541	INR
Bending die (V-block)	x0	2,541	INR
Pressure pad · quantity	op_bend + op_draw	1	pcs
Pressure pad · weight each	die_l * die_w * 20 * density / 1e6	9.3423	kg
Pressure pad · steel weight	qty * weight	9.3423	kg
Pressure pad · steel C45 / S50C	qty * weight * rate	1,028	INR
Pressure pad	x0	1,028	INR
Stock guides & stopper · quantity	1	1	pcs
Stock guides & stopper · weight each	2 * die_l * 30 * 15 * density / 1e6	2.8027	kg
Stock guides & stopper · steel weight	qty * weight	2.8027	kg
Stock guides & stopper · steel C45 / S50C	qty * weight * rate	308.2954	INR
Stock guides & stopper	x0	308.2954	INR
Die springs / urethane · quantity	4 + 2 * ceil(die_l / 200)	8	pcs
Die springs / urethane · catalogue price each	450	450	INR
Die springs / urethane · bought-out	qty * each	3,600	INR

Step	Formula	Value	Unit
Die springs / urethane	x0	3,600	INR
Screws, dowels & bolts (set) · quantity	1	1	pcs
Screws, dowels & bolts (set) · catalogue price each	1500 + die_l * die_w / 100	2,095	INR
Screws, dowels & bolts (set) · bought-out	qty * each	2,095	INR
Screws, dowels & bolts (set)	x0	2,095	INR
CNC machining - plates & forming steels · quantity	1	1	pcs
CNC machining - plates & forming steels · hours each	cnc_setup_h + 4 * die_l * die_w / 100 / cnc_plate_cm2_per_h	7.9505	h
CNC machining - plates & forming steels · hours	qty * hours	7.9505	h
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	450,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	405,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	105.4688	INR/h
Average investment	(machine_price + residual_value) / 2	2,475,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	64.4531	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	76.5	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	46.875	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	13.125	INR/h
Consumables & tooling	consumables_rate	150	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	456.4219	INR/h
Overhead	machine_cost * overhead_pct / 100	68.4633	INR/h
MHR VMC-850 · CNC vertical machining centre 850	machine_cost + overhead	524.8852	INR/h
Machine hour rate (MHR) in INR	value * fx	524.8852	INR/h
Paid hours per year	working_days * hours_per_shift	2,400	h/yr
Annual salary / CTC	monthly_salary * 12	360,000	INR/yr
Employer extra cost	annual_salary * employer_extra_pct / 100	0	INR/yr
Base rate	(annual_salary + employer_cost) / paid_hours_per_year	150	INR/h
Supervision & labour overhead	base_rate * supervision_pct / 100	22.5	INR/h

Step	Formula	Value	Unit
Non-productive time	(base_rate + supervision) * (100 / productive_pct - 1)	30.4412	INR/h
Labour rate · skilled_machinist	base_rate + supervision + productivity_loss	202.9412	INR/h
Labour rate in INR	value * fx	202.9412	INR/h
CNC milling: cost per hour	mhr + operators * labour	727.8263	INR/h
CNC machining - plates & forming steels · CNC milling	hours * cost_per_hour	5,787	INR
CNC machining - plates & forming steels	x0	5,787	INR
Surface grinding · quantity	1	1	pcs
Surface grinding · hours each	grind_setup_h + 6 * die_l * die_w / 100 / grind_cm2_per_h	12.901	h
Surface grinding · hours	qty * hours	12.901	h
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	90,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	142,500	INR/yr
Depreciation	depreciation_per_year / annual_hours	37.1094	INR/h
Average investment	(machine_price + residual_value) / 2	945,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	24.6094	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	38.25	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	14.0625	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	8.75	INR/h
Consumables & tooling	consumables_rate	80	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	202.7812	INR/h
Overhead	machine_cost * overhead_pct / 100	30.4172	INR/h
MHR GRIND-SURF · Surface grinder	machine_cost + overhead	233.1984	INR/h
Machine hour rate (MHR) in INR	value * fx	233.1984	INR/h
Grinding: cost per hour	mhr + operators * labour	436.1396	INR/h
Surface grinding · Grinding	hours * cost_per_hour	5,627	INR
Surface grinding	x0	5,627	INR
Heat treatment (die steels) · quantity	1	1	pcs

Step	Formula	Value	Unit
Die plate height	max(die_plate_min_mm, die_plate_t_factor * t)	25	mm
Heat treatment (die steels) · amount each	die_l * die_w * (die_h + 2 * backing_h_mm) * 7.85 / 1e6 + (op_blank + op_trim) * blank_l * blank_w * punch_len_mm * 7.85 / 1e6	22.8886	kg
Heat treatment (die steels) · Heat treatment (per kg)	max(min_charge, qty * amount * rate)	2,747	INR
Heat treatment (die steels)	x0	2,747	INR
Fitting & assembly · quantity	1	1	pcs
Cutting punches in this die	op_blank * 1 + op_pierce * hole_count + op_trim * 1	0	pcs
Fitting & assembly · hours each	assy_h_base + assy_h_per_punch * punches	16	h
Fitting & assembly · hours	qty * hours	16	h
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	350,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	210,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	54.6875	INR/h
Average investment	(machine_price + residual_value) / 2	1,925,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	50.1302	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	19.125	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	18.2292	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	16.4062	INR/h
Consumables & tooling	consumables_rate	30	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	188.5781	INR/h
Overhead	machine_cost * overhead_pct / 100	28.2867	INR/h
MHR SPOTTING · Spotting / try-out press	machine_cost + overhead	216.8648	INR/h
Machine hour rate (MHR) in INR	value * fx	216.8648	INR/h
Paid hours per year	working_days * hours_per_shift	2,400	h/yr
Annual salary / CTC	monthly_salary * 12	540,000	INR/yr
Employer extra cost	annual_salary * employer_extra_pct / 100	0	INR/yr
Base rate	(annual_salary + employer_cost) / paid_hours_per_year	225	INR/h
Supervision & labour overhead	base_rate * supervision_pct / 100	33.75	INR/h

Step	Formula	Value	Unit
Non-productive time	(base_rate + supervision) * (100 / productive_pct - 1)	45.6618	INR/h
Labour rate · toolmaker	base_rate + supervision + productivity_loss	304.4118	INR/h
Labour rate in INR	value * fx	304.4118	INR/h
Fitting & assembly: cost per hour	mhr + operators * labour	825.6884	INR/h
Fitting & assembly · Fitting & assembly	hours * cost_per_hour	13,211	INR
Fitting & assembly	x0	13,211	INR
Die design · quantity	1	1	pcs
Die design · hours each	design_h_base + design_h_per_op * die_ops	18	h
Die design · hours	qty * hours	18	h
Paid hours per year	working_days * hours_per_shift	2,400	h/yr
Annual salary / CTC	monthly_salary * 12	840,000	INR/yr
Employer extra cost	annual_salary * employer_extra_pct / 100	0	INR/yr
Base rate	(annual_salary + employer_cost) / paid_hours_per_year	350	INR/h
Supervision & labour overhead	base_rate * supervision_pct / 100	52.5	INR/h
Non-productive time	(base_rate + supervision) * (100 / productive_pct - 1)	71.0294	INR/h
Labour rate · designer	base_rate + supervision + productivity_loss	473.5294	INR/h
Labour rate in INR	value * fx	473.5294	INR/h
Design: cost per hour	operators * labour	473.5294	INR/h
Die design · Design	hours * cost_per_hour	8,524	INR
Die design	x0	8,524	INR
Machining (CNC, EDM, wire cut, grinding, drilling)	x0 + x1	11,413	INR
Heat & surface treatment	x0	2,747	INR
Fitting & assembly	x0	13,211	INR
Design	x0	8,524	INR
Tool steel	x0 + x1 + x2 + x3 + x4 + x5	7,070	INR
Mould base & standard parts	x0 + x1 + x2	73,695	INR
Processes (design excluded)	x0 + x1 + x2	27,371	INR

Step	Formula	Value	Unit
Design	x0	8,524	INR
Tool weight	x0 + x1 + x2 + x3 + x4 + x5 + x6	299.6998	kg
Shot weight (net)	cavities * part_weight_g / 1000	0.1998	kg
Press force needed for this die	die_force_t * safety	0.6442	t
Try-out machine: PRESS-100 · Power press 100 t	smallest 'Press' machine in Master data → Machines whose tonnage ≥ the press force	100	t
Machine hours per year	working_days * shifts_per_day * hours_per_shift * utilisation_pct / 100	3,840	h/yr
Residual value	machine_price * residual_pct / 100	300,000	INR
Depreciation per year	(machine_price - residual_value) / life_years	180,000	INR/yr
Depreciation	depreciation_per_year / annual_hours	46.875	INR/h
Average investment	(machine_price + residual_value) / 2	1,650,000	INR
Interest	average_investment * interest_pct / 100 / annual_hours	42.9688	INR/h
Power	power_kw * load_factor_pct / 100 * electricity_rate	56.1	INR/h
Maintenance	machine_price * maintenance_pct / 100 / annual_hours	31.25	INR/h
Floor space	floor_area_m2 * rent_per_m2_month * 12 / annual_hours	8.75	INR/h
Consumables & tooling	consumables_rate	60	INR/h
Machine cost subtotal	depreciation + interest + power + maintenance + floor_space + consumables	245.9437	INR/h
Overhead	machine_cost * overhead_pct / 100	36.8916	INR/h
MHR PRESS-100 · Power press 100 t	machine_cost + overhead	282.8353	INR/h
Machine hour rate (MHR) in INR	value * fx	282.8353	INR/h
Try-out crew per hour (operators × labour rate)	operators * labour	304.4118	INR/h
Try-out hours	trials * trial_hours_each	6	h
Try-out machine & crew	trial_hours * (trial_mhr + trial_labour_per_h)	3,523	INR
Try-out material	trials * trial_shots_each * shot_weight_kg * trial_material_rate	5,755	INR
Try-out	trial_machine_cost + trial_material_cost	9,279	INR
Manufacturing cost	material_cost + bought_out_cost + process_cost + design_cost + trial_cost	125,938	INR
Toolmaker overhead	manufacturing_cost * overhead_pct / 100	15,113	INR
Toolmaker profit	(manufacturing_cost + tool_overhead) * profit_pct / 100	14,105	INR

Step	Formula	Value	Unit
Tool price ex-works	manufacturing_cost + tool_overhead + tool_profit	155,155	INR
Packing	tool_price * packing_pct / 100	1,552	INR
Transport	tool_weight_kg * transport_rate	2,398	INR
Warranty / spares	tool_price * warranty_pct / 100 * warranty_on	4,655	INR
Die 3: Bending (V) · tool cost	tool_price + tool_packing + tool_transport + tool_warranty	163,759	INR
Die 3: Bending (V) · tool CAPEX	tool_cost * tools_count	163,759	INR
Tool cost per tool in INR	value * fx	163,759	INR
Tool CAPEX in INR	value * fx	163,759	INR
Die set: tool cost of all dies	d0 + d1 + d2	530,911	INR
Tool CAPEX: all dies × tools over the programme	d0 + d1 + d2	530,911	INR
Die set tool cost in INR	value * fx	530,911	INR
Tool CAPEX in INR	value * fx	530,911	INR
Machining (CNC, EDM, wire cut, grinding, drilling) (all dies)	d0 + d1 + d2	50,453	INR
Heat & surface treatment (all dies)	d0 + d1 + d2	8,978	INR
Fitting & assembly (all dies)	d0 + d1 + d2	42,523	INR
Design (all dies)	d0 + d1 + d2	25,571	INR
Tool steel (all dies)	d0 + d1 + d2	29,615	INR
Die sets & standard parts (all dies)	d0 + d1 + d2	223,665	INR
Processes (design excluded) (all dies)	d0 + d1 + d2	101,955	INR
Design (all dies)	d0 + d1 + d2	25,571	INR
Weight of all dies	d0 + d1 + d2	915.9111	kg

Inputs and where each comes from

Input	Value	Unit	Comes from
Density of C45 / S50C	7.85	g/cm³	Master data → Mould & die database → Tool materials · C45 / S50C · EXAMPLE
Mould base weight	260	kg	Master data → Mould & die database → Mould bases · 4-pillar die set 5040 · EXAMPLE
Steel rate · C45 / S50C	110	INR/kg	Rate library #598 · tool_material / C45 / S50C · IN · 2026-10 · approved · EXAMPLE

Input	Value	Unit	Comes from
Die set 4-pillar die set 5040 · C45 / S50C · catalogue price	68,000	INR	Master data → Mould & die database → Mould bases · 4-pillar die set 5040 · EXAMPLE
Sheet thickness	2	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Annual volume	200,000	parts/yr	Part register · Stage C for 200,000 parts/yr (Σ variant annual volume × quantity (STD 200,000 × 1)), 5 programme years
Programme years	5	years	Part register · Stage C for 200,000 parts/yr (Σ variant annual volume × quantity (STD 200,000 × 1)), 5 programme years
Cavities per tool	1	pcs	Your input · Tool features
Density of D2 (1.2379)	7.7	g/cm³	Master data → Mould & die database → Tool materials · D2 (1.2379) · EXAMPLE
Smallest die plate height	25	mm	Master data → Mould & die database → Sizing rules · sheet_metal / die_plate_min_mm · EXAMPLE
Die plate height as a multiple of the sheet thickness	12	× t	Master data → Mould & die database → Sizing rules · sheet_metal / die_plate_t_factor · EXAMPLE
Die work area length	326.7	mm	Part register · Die 1: Blanking: blank / part envelope 326.7 × 80.0 mm
Die plate steel around the work area (blank or strip), each side	35	mm	Master data → Mould & die database → Sizing rules · sheet_metal / die_wall_mm · EXAMPLE
Die work area width	80	mm	Part register · Die 1: Blanking: blank / part envelope 326.7 × 80.0 mm
Steel rate · D2 (1.2379)	500	INR/kg	Rate library #599 · tool_material / D2 (1.2379) · IN · 2026-10 · approved · EXAMPLE
blank operations in this die	1		Part register · Die 1: Blanking: operations blank (suggested die split)
trim operations in this die	0		Part register · Die 1: Blanking: operations blank (suggested die split)
Blank length	326.7	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Punch length	60	mm	Master data → Mould & die database → Sizing rules · sheet_metal / punch_len_mm · EXAMPLE
Blank width	40	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Punch holder plate thickness	22	mm	Master data → Mould & die database → Sizing rules · sheet_metal / punch_plate_h_mm · EXAMPLE
Stripper plate thickness	20	mm	Master data → Mould & die database → Sizing rules · sheet_metal / stripper_h_mm · EXAMPLE
Density of Mould base steel (4140 / 1.7225)	7.85	g/cm³	Master data → Mould & die database → Tool materials · Mould base steel (4140 / 1.7225) · EXAMPLE
Hardened backing plate thickness	12	mm	Master data → Mould & die database → Sizing rules · sheet_metal / backing_h_mm · EXAMPLE
Steel rate · Mould base steel (4140 / 1.7225)	140	INR/kg	Rate library #602 · tool_material / Mould base steel (4140 / 1.7225) · IN · 2026-10 · approved · EXAMPLE
Progressive die (1 = yes)	0		Part register · Die 1: Blanking: blank / part envelope 326.7 × 80.0 mm
Stations	1		Part register · Die 1: Blanking: blank / part envelope 326.7 × 80.0 mm

Input	Value	Unit	Comes from
Blank perimeter	733.4	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
pierce operations in this die	0		Part register · Die 1: Blanking: operations blank (suggested die split)
Total hole perimeter	160.2212	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Wire cut set-up per profile	1	h	Master data → Mould & die database → Process time parameters · sheet_metal / wirecut_setup_h · EXAMPLE
Wire cut speed: cut length × height per hour in hardened die steel	4,800	mm²/h	Master data → Mould & die database → Process time parameters · sheet_metal / wirecut_mm2_per_h · EXAMPLE
Machine price	6,000,000	INR	Rate library #33 · machine_price / WEDM · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	250	INR/h	Rate library #34 · consumables / WEDM · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	10	kW	Master data → Machines · WEDM · power_kw
Power load factor	70	%	Master data → Machines · WEDM · load_factor_pct
Floor area	8	m²	Master data → Machines · WEDM · floor_area_m2
Machine life	10	years	Master data → Machines · WEDM · life_years
Residual value	10	%	Master data → Machines · WEDM · residual_pct
Maintenance per year	5	% of price	Master data → Machines · WEDM · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Wire cut	0.5	persons	Master data → Mould & die database → Tool processes · Wire cut · EXAMPLE
Monthly salary / CTC	30,000	INR/month	Rate library #22 · labour_salary / skilled_machinist · IN · 2026-10 · approved · EXAMPLE
Employer extra cost	0	%	Rate library #18 · labour / employer_extra_cost_pct · IN · 2026-10 · approved · EXAMPLE
Supervision & labour overhead	15	%	Rate library #19 · labour / supervision_pct · IN · 2026-10 · approved · EXAMPLE

Input	Value	Unit	Comes from
Productive time	85	%	Rate library #20 · labour / productive_time_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
CNC: plate area machined per hour (pockets, holes, threads)	400	cm²/h	Master data → Mould & die database → Process time parameters · sheet_metal / cnc_plate_cm2_per_h · EXAMPLE
CNC set-up per die	2	h	Master data → Mould & die database → Process time parameters · sheet_metal / cnc_setup_h · EXAMPLE
Machine price	4,500,000	INR	Rate library #27 · machine_price / VMC-850 · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	150	INR/h	Rate library #28 · consumables / VMC-850 · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	15	kW	Master data → Machines · VMC-850 · power_kw
Power load factor	60	%	Master data → Machines · VMC-850 · load_factor_pct
Floor area	12	m²	Master data → Machines · VMC-850 · floor_area_m2
Machine life	10	years	Master data → Machines · VMC-850 · life_years
Residual value	10	%	Master data → Machines · VMC-850 · residual_pct
Maintenance per year	4	% of price	Master data → Machines · VMC-850 · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · CNC milling	1	persons	Master data → Mould & die database → Tool processes · CNC milling · EXAMPLE
Grinding set-up per die	1	h	Master data → Mould & die database → Process time parameters · sheet_metal / grind_setup_h · EXAMPLE
Surface grinding area per hour	300	cm²/h	Master data → Mould & die database → Process time parameters · sheet_metal / grind_cm2_per_h · EXAMPLE

Input	Value	Unit	Comes from
Machine price	1,800,000	INR	Rate library #37 · machine_price / GRIND-SURF · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	80	INR/h	Rate library #38 · consumables / GRIND-SURF · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	7.5	kW	Master data → Machines · GRIND-SURF · power_kw
Power load factor	60	%	Master data → Machines · GRIND-SURF · load_factor_pct
Floor area	8	m²	Master data → Machines · GRIND-SURF · floor_area_m2
Machine life	12	years	Master data → Machines · GRIND-SURF · life_years
Residual value	5	%	Master data → Machines · GRIND-SURF · residual_pct
Maintenance per year	3	% of price	Master data → Machines · GRIND-SURF · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Grinding	1	persons	Master data → Mould & die database → Tool processes · Grinding · EXAMPLE
Rate · Heat treatment (per kg)	120	INR/kg	Rate library #613 · process_unit / Heat treatment (per kg) · IN · 2026-10 · approved · EXAMPLE
Minimum charge per job · Heat treatment (per kg)	1,500	INR/job	Rate library #614 · process_min_charge / Heat treatment (per kg) · IN · 2026-10 · approved · EXAMPLE
Fitting per cutting punch	0.5	h	Master data → Mould & die database → Process time parameters · sheet_metal / assy_h_per_punch · EXAMPLE
Fitting & assembly of one die	16	h	Master data → Mould & die database → Process time parameters · sheet_metal / assy_h_base · EXAMPLE
Pierced holes	6	pcs	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Machine price	3,500,000	INR	Rate library #43 · machine_price / SPOTTING · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	30	INR/h	Rate library #44 · consumables / SPOTTING · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE

Input	Value	Unit	Comes from
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	7.5	kW	Master data → Machines · SPOTTING · power_kw
Power load factor	30	%	Master data → Machines · SPOTTING · load_factor_pct
Floor area	15	m²	Master data → Machines · SPOTTING · floor_area_m2
Machine life	15	years	Master data → Machines · SPOTTING · life_years
Residual value	10	%	Master data → Machines · SPOTTING · residual_pct
Maintenance per year	2	% of price	Master data → Machines · SPOTTING · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Fitting & assembly	2	persons	Master data → Mould & die database → Tool processes · Fitting & assembly · EXAMPLE
Monthly salary / CTC	45,000	INR/month	Rate library #23 · labour_salary / toolmaker · IN · 2026-10 · approved · EXAMPLE
Employer extra cost	0	%	Rate library #18 · labour / employer_extra_cost_pct · IN · 2026-10 · approved · EXAMPLE
Supervision & labour overhead	15	%	Rate library #19 · labour / supervision_pct · IN · 2026-10 · approved · EXAMPLE
Productive time	85	%	Rate library #20 · labour / productive_time_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
Die design per operation / station	6	h	Master data → Mould & die database → Process time parameters · sheet_metal / design_h_per_op · EXAMPLE
Operations in this die	1		Part register · Die 1: Blanking: blank / part envelope 326.7 × 80.0 mm
Die design per die	12	h	Master data → Mould & die database → Process time parameters · sheet_metal / design_h_base · EXAMPLE
Operators · Design	1	persons	Master data → Mould & die database → Tool processes · Design · EXAMPLE

Input	Value	Unit	Comes from
Monthly salary / CTC	70,000	INR/month	Rate library #24 · labour_salary / designer · IN · 2026-10 · approved · EXAMPLE
Employer extra cost	0	%	Rate library #18 · labour / employer_extra_cost_pct · IN · 2026-10 · approved · EXAMPLE
Supervision & labour overhead	15	%	Rate library #19 · labour / supervision_pct · IN · 2026-10 · approved · EXAMPLE
Productive time	85	%	Rate library #20 · labour / productive_time_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
Net weight of one part	199.8304	g	Part register · Net weight = CAD volume × material density (Materials)
Warranty / spares included (1 = yes)	1		Part register · Default: included
Tools over the programme	1	tools	Part register · Stage C for 200,000 parts/yr (Σ variant annual volume × quantity (STD 200,000 × 1)), 5 programme years: max(tools for capacity 1, tools for tool life 1)
Projected area of one part	0	cm²	Your input · Estimate page
Press safety factor	1.25	×	Master data → Parameters · sheet_metal / press_safety_factor
Force of this die (Σ its operations)	49.3588	t	Part register · Die 1: Blanking: blank / part envelope 326.7 × 80.0 mm
Machine price	3,000,000	INR	Rate library #63 · machine_price / PRESS-100 · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	60	INR/h	Rate library #64 · consumables / PRESS-100 · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	11	kW	Master data → Machines · PRESS-100 · power_kw
Power load factor	60	%	Master data → Machines · PRESS-100 · load_factor_pct
Floor area	8	m²	Master data → Machines · PRESS-100 · floor_area_m2
Machine life	15	years	Master data → Machines · PRESS-100 · life_years

Input	Value	Unit	Comes from
Residual value	10	%	Master data → Machines · PRESS-100 · residual_pct
Maintenance per year	4	% of price	Master data → Machines · PRESS-100 · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Tryout	1	persons	Master data → Mould & die database → Tool processes · Tryout · EXAMPLE
Number of trials	2	trials	Master data → Parameters · sheet_metal / trials
Machine hours per trial	3	h	Master data → Parameters · sheet_metal / trial_hours_each
Shots per trial	200	shots	Master data → Parameters · sheet_metal / trial_shots_each
Part material rate (trial shots)	72	INR/kg	Rate library #585 · material / Galvanised steel GI · IN · 2026-10 · approved · EXAMPLE
Toolmaker overhead	12	%	Rate library #1341 · tool_commercial / tool_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Toolmaker profit	10	%	Rate library #1342 · tool_commercial / tool_profit_pct · IN · 2026-10 · approved · EXAMPLE
Packing	1	%	Rate library #1343 · tool_commercial / tool_packing_pct · IN · 2026-10 · approved · EXAMPLE
Warranty / spares	3	%	Rate library #1344 · tool_commercial / tool_warranty_pct · IN · 2026-10 · approved · EXAMPLE
Transport per kg	8	INR/kg	Rate library #635 · process_unit / Transport (per kg) · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
Density of C45 / S50C	7.85	g/cm³	Master data → Mould & die database → Tool materials · C45 / S50C · EXAMPLE
Mould base weight	260	kg	Master data → Mould & die database → Mould bases · 4-pillar die set 5040 · EXAMPLE
Steel rate · C45 / S50C	110	INR/kg	Rate library #598 · tool_material / C45 / S50C · IN · 2026-10 · approved · EXAMPLE
Die set 4-pillar die set 5040 · C45 / S50C · catalogue price	68,000	INR	Master data → Mould & die database → Mould bases · 4-pillar die set 5040 · EXAMPLE
Sheet thickness	2	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Annual volume	200,000	parts/yr	Part register · Stage C for 200,000 parts/yr (Σ variant annual volume × quantity (STD 200,000 × 1)), 5 programme years
Programme years	5	years	Part register · Stage C for 200,000 parts/yr (Σ variant annual volume × quantity (STD 200,000 × 1)), 5 programme years
Cavities per tool	1	pcs	Your input · Tool features
Density of D2 (1.2379)	7.7	g/cm³	Master data → Mould & die database → Tool materials · D2 (1.2379) · EXAMPLE
Smallest die plate height	25	mm	Master data → Mould & die database → Sizing rules · sheet_metal / die_plate_min_mm · EXAMPLE
Die plate height as a multiple of the sheet thickness	12	× t	Master data → Mould & die database → Sizing rules · sheet_metal / die_plate_t_factor · EXAMPLE

Input	Value	Unit	Comes from
Die work area length	326.7	mm	Part register · Die 2: Piercing: blank / part envelope 326.7 × 80.0 mm
Die plate steel around the work area (blank or strip), each side	35	mm	Master data → Mould & die database → Sizing rules · sheet_metal / die_wall_mm · EXAMPLE
Die work area width	80	mm	Part register · Die 2: Piercing: blank / part envelope 326.7 × 80.0 mm
Steel rate · D2 (1.2379)	500	INR/kg	Rate library #599 · tool_material / D2 (1.2379) · IN · 2026-10 · approved · EXAMPLE
Pierced holes	6	pcs	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
pierce operations in this die	1		Part register · Die 2: Piercing: operations pierce (suggested die split)
Punch holder plate thickness	22	mm	Master data → Mould & die database → Sizing rules · sheet_metal / punch_plate_h_mm · EXAMPLE
Stripper plate thickness	20	mm	Master data → Mould & die database → Sizing rules · sheet_metal / stripper_h_mm · EXAMPLE
Density of Mould base steel (4140 / 1.7225)	7.85	g/cm³	Master data → Mould & die database → Tool materials · Mould base steel (4140 / 1.7225) · EXAMPLE
Hardened backing plate thickness	12	mm	Master data → Mould & die database → Sizing rules · sheet_metal / backing_h_mm · EXAMPLE
Steel rate · Mould base steel (4140 / 1.7225)	140	INR/kg	Rate library #602 · tool_material / Mould base steel (4140 / 1.7225) · IN · 2026-10 · approved · EXAMPLE
Progressive die (1 = yes)	0		Part register · Die 2: Piercing: blank / part envelope 326.7 × 80.0 mm
Stations	1		Part register · Die 2: Piercing: blank / part envelope 326.7 × 80.0 mm
blank operations in this die	0		Part register · Die 2: Piercing: operations pierce (suggested die split)
Blank perimeter	733.4	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Total hole perimeter	160.2212	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
trim operations in this die	0		Part register · Die 2: Piercing: operations pierce (suggested die split)
Wire cut set-up per profile	1	h	Master data → Mould & die database → Process time parameters · sheet_metal / wirecut_setup_h · EXAMPLE
Wire cut speed: cut length × height per hour in hardened die steel	4,800	mm²/h	Master data → Mould & die database → Process time parameters · sheet_metal / wirecut_mm2_per_h · EXAMPLE
Machine price	6,000,000	INR	Rate library #33 · machine_price / WEDM · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	250	INR/h	Rate library #34 · consumables / WEDM · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE

Input	Value	Unit	Comes from
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	10	kW	Master data → Machines · WEDM · power_kw
Power load factor	70	%	Master data → Machines · WEDM · load_factor_pct
Floor area	8	m²	Master data → Machines · WEDM · floor_area_m2
Machine life	10	years	Master data → Machines · WEDM · life_years
Residual value	10	%	Master data → Machines · WEDM · residual_pct
Maintenance per year	5	% of price	Master data → Machines · WEDM · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Wire cut	0.5	persons	Master data → Mould & die database → Tool processes · Wire cut · EXAMPLE
Monthly salary / CTC	30,000	INR/month	Rate library #22 · labour_salary / skilled_machinist · IN · 2026-10 · approved · EXAMPLE
Employer extra cost	0	%	Rate library #18 · labour / employer_extra_cost_pct · IN · 2026-10 · approved · EXAMPLE
Supervision & labour overhead	15	%	Rate library #19 · labour / supervision_pct · IN · 2026-10 · approved · EXAMPLE
Productive time	85	%	Rate library #20 · labour / productive_time_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
Punch length	60	mm	Master data → Mould & die database → Sizing rules · sheet_metal / punch_len_mm · EXAMPLE
CNC: plate area machined per hour (pockets, holes, threads)	400	cm²/h	Master data → Mould & die database → Process time parameters · sheet_metal / cnc_plate_cm2_per_h · EXAMPLE
CNC set-up per die	2	h	Master data → Mould & die database → Process time parameters · sheet_metal / cnc_setup_h · EXAMPLE
Machine price	4,500,000	INR	Rate library #27 · machine_price / VMC-850 · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	150	INR/h	Rate library #28 · consumables / VMC-850 · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE

Input	Value	Unit	Comes from
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	15	kW	Master data → Machines · VMC-850 · power_kw
Power load factor	60	%	Master data → Machines · VMC-850 · load_factor_pct
Floor area	12	m²	Master data → Machines · VMC-850 · floor_area_m2
Machine life	10	years	Master data → Machines · VMC-850 · life_years
Residual value	10	%	Master data → Machines · VMC-850 · residual_pct
Maintenance per year	4	% of price	Master data → Machines · VMC-850 · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · CNC milling	1	persons	Master data → Mould & die database → Tool processes · CNC milling · EXAMPLE
Grinding set-up per die	1	h	Master data → Mould & die database → Process time parameters · sheet_metal / grind_setup_h · EXAMPLE
Surface grinding area per hour	300	cm²/h	Master data → Mould & die database → Process time parameters · sheet_metal / grind_cm2_per_h · EXAMPLE
Machine price	1,800,000	INR	Rate library #37 · machine_price / GRIND-SURF · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	80	INR/h	Rate library #38 · consumables / GRIND-SURF · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE

Input	Value	Unit	Comes from
Connected power	7.5	kW	Master data → Machines · GRIND-SURF · power_kw
Power load factor	60	%	Master data → Machines · GRIND-SURF · load_factor_pct
Floor area	8	m²	Master data → Machines · GRIND-SURF · floor_area_m2
Machine life	12	years	Master data → Machines · GRIND-SURF · life_years
Residual value	5	%	Master data → Machines · GRIND-SURF · residual_pct
Maintenance per year	3	% of price	Master data → Machines · GRIND-SURF · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Grinding	1	persons	Master data → Mould & die database → Tool processes · Grinding · EXAMPLE
Blank width	40	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Blank length	326.7	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Rate · Heat treatment (per kg)	120	INR/kg	Rate library #613 · process_unit / Heat treatment (per kg) · IN · 2026-10 · approved · EXAMPLE
Minimum charge per job · Heat treatment (per kg)	1,500	INR/job	Rate library #614 · process_min_charge / Heat treatment (per kg) · IN · 2026-10 · approved · EXAMPLE
Fitting per cutting punch	0.5	h	Master data → Mould & die database → Process time parameters · sheet_metal / assy_h_per_punch · EXAMPLE
Fitting & assembly of one die	16	h	Master data → Mould & die database → Process time parameters · sheet_metal / assy_h_base · EXAMPLE
Machine price	3,500,000	INR	Rate library #43 · machine_price / SPOTTING · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	30	INR/h	Rate library #44 · consumables / SPOTTING · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	7.5	kW	Master data → Machines · SPOTTING · power_kw
Power load factor	30	%	Master data → Machines · SPOTTING · load_factor_pct

Input	Value	Unit	Comes from
Floor area	15	m²	Master data → Machines · SPOTTING · floor_area_m2
Machine life	15	years	Master data → Machines · SPOTTING · life_years
Residual value	10	%	Master data → Machines · SPOTTING · residual_pct
Maintenance per year	2	% of price	Master data → Machines · SPOTTING · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Fitting & assembly	2	persons	Master data → Mould & die database → Tool processes · Fitting & assembly · EXAMPLE
Monthly salary / CTC	45,000	INR/month	Rate library #23 · labour_salary / toolmaker · IN · 2026-10 · approved · EXAMPLE
Employer extra cost	0	%	Rate library #18 · labour / employer_extra_cost_pct · IN · 2026-10 · approved · EXAMPLE
Supervision & labour overhead	15	%	Rate library #19 · labour / supervision_pct · IN · 2026-10 · approved · EXAMPLE
Productive time	85	%	Rate library #20 · labour / productive_time_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
Die design per operation / station	6	h	Master data → Mould & die database → Process time parameters · sheet_metal / design_h_per_op · EXAMPLE
Operations in this die	1		Part register · Die 2: Piercing: blank / part envelope 326.7 × 80.0 mm
Die design per die	12	h	Master data → Mould & die database → Process time parameters · sheet_metal / design_h_base · EXAMPLE
Operators · Design	1	persons	Master data → Mould & die database → Tool processes · Design · EXAMPLE
Monthly salary / CTC	70,000	INR/month	Rate library #24 · labour_salary / designer · IN · 2026-10 · approved · EXAMPLE
Employer extra cost	0	%	Rate library #18 · labour / employer_extra_cost_pct · IN · 2026-10 · approved · EXAMPLE
Supervision & labour overhead	15	%	Rate library #19 · labour / supervision_pct · IN · 2026-10 · approved · EXAMPLE
Productive time	85	%	Rate library #20 · labour / productive_time_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
Net weight of one part	199.8304	g	Part register · Net weight = CAD volume × material density (Materials)
Warranty / spares included (1 = yes)	1		Part register · Default: included

Input	Value	Unit	Comes from
Tools over the programme	1	tools	Part register · Stage C for 200,000 parts/yr (Σ variant annual volume × quantity (STD 200,000 × 1)), 5 programme years: max(tools for capacity 1, tools for tool life 1)
Projected area of one part	0	cm²	Your input · Estimate page
Press safety factor	1.25	×	Master data → Parameters · sheet_metal / press_safety_factor
Force of this die (Σ its operations)	10.7831	t	Part register · Die 2: Piercing: blank / part envelope 326.7 × 80.0 mm
Machine price	3,000,000	INR	Rate library #63 · machine_price / PRESS-100 · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	60	INR/h	Rate library #64 · consumables / PRESS-100 · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	11	kW	Master data → Machines · PRESS-100 · power_kw
Power load factor	60	%	Master data → Machines · PRESS-100 · load_factor_pct
Floor area	8	m²	Master data → Machines · PRESS-100 · floor_area_m2
Machine life	15	years	Master data → Machines · PRESS-100 · life_years
Residual value	10	%	Master data → Machines · PRESS-100 · residual_pct
Maintenance per year	4	% of price	Master data → Machines · PRESS-100 · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Tryout	1	persons	Master data → Mould & die database → Tool processes · Tryout · EXAMPLE
Number of trials	2	trials	Master data → Parameters · sheet_metal / trials
Machine hours per trial	3	h	Master data → Parameters · sheet_metal / trial_hours_each
Shots per trial	200	shots	Master data → Parameters · sheet_metal / trial_shots_each
Part material rate (trial shots)	72	INR/kg	Rate library #585 · material / Galvanised steel GI · IN · 2026-10 · approved · EXAMPLE
Toolmaker overhead	12	%	Rate library #1341 · tool_commercial / tool_overhead_pct · IN · 2026-10 · approved · EXAMPLE

Input	Value	Unit	Comes from
Toolmaker profit	10	%	Rate library #1342 · tool_commercial / tool_profit_pct · IN · 2026-10 · approved · EXAMPLE
Packing	1	%	Rate library #1343 · tool_commercial / tool_packing_pct · IN · 2026-10 · approved · EXAMPLE
Warranty / spares	3	%	Rate library #1344 · tool_commercial / tool_warranty_pct · IN · 2026-10 · approved · EXAMPLE
Transport per kg	8	INR/kg	Rate library #635 · process_unit / Transport (per kg) · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
Density of C45 / S50C	7.85	g/cm³	Master data → Mould & die database → Tool materials · C45 / S50C · EXAMPLE
Mould base weight	260	kg	Master data → Mould & die database → Mould bases · 4-pillar die set 5040 · EXAMPLE
Steel rate · C45 / S50C	110	INR/kg	Rate library #598 · tool_material / C45 / S50C · IN · 2026-10 · approved · EXAMPLE
Die set 4-pillar die set 5040 · C45 / S50C · catalogue price	68,000	INR	Master data → Mould & die database → Mould bases · 4-pillar die set 5040 · EXAMPLE
Die work area length	326.7	mm	Part register · Die 3: Bending (V): blank / part envelope 326.7 × 80.0 mm
Die plate steel around the work area (blank or strip), each side	35	mm	Master data → Mould & die database → Sizing rules · sheet_metal / die_wall_mm · EXAMPLE
Punch holder plate thickness	22	mm	Master data → Mould & die database → Sizing rules · sheet_metal / punch_plate_h_mm · EXAMPLE
Die work area width	80	mm	Part register · Die 3: Bending (V): blank / part envelope 326.7 × 80.0 mm
Density of Mould base steel (4140 / 1.7225)	7.85	g/cm³	Master data → Mould & die database → Tool materials · Mould base steel (4140 / 1.7225) · EXAMPLE
Hardened backing plate thickness	12	mm	Master data → Mould & die database → Sizing rules · sheet_metal / backing_h_mm · EXAMPLE
Steel rate · Mould base steel (4140 / 1.7225)	140	INR/kg	Rate library #602 · tool_material / Mould base steel (4140 / 1.7225) · IN · 2026-10 · approved · EXAMPLE
bend operations in this die	1		Part register · Die 3: Bending (V): operations bend (suggested die split)
Density of D2 (1.2379)	7.7	g/cm³	Master data → Mould & die database → Tool materials · D2 (1.2379) · EXAMPLE
Sheet thickness	2	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Blank width	40	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Steel rate · D2 (1.2379)	500	INR/kg	Rate library #599 · tool_material / D2 (1.2379) · IN · 2026-10 · approved · EXAMPLE
draw operations in this die	0		Part register · Die 3: Bending (V): operations bend (suggested die split)
Progressive die (1 = yes)	0		Part register · Die 3: Bending (V): blank / part envelope 326.7 × 80.0 mm
Stations	1		Part register · Die 3: Bending (V): blank / part envelope 326.7 × 80.0 mm
CNC: plate area machined per hour (pockets, holes, threads)	400	cm²/h	Master data → Mould & die database → Process time parameters · sheet_metal / cnc_plate_cm2_per_h · EXAMPLE

Input	Value	Unit	Comes from
CNC set-up per die	2	h	Master data → Mould & die database → Process time parameters · sheet_metal / cnc_setup_h · EXAMPLE
Machine price	4,500,000	INR	Rate library #27 · machine_price / VMC-850 · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	150	INR/h	Rate library #28 · consumables / VMC-850 · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	15	kW	Master data → Machines · VMC-850 · power_kw
Power load factor	60	%	Master data → Machines · VMC-850 · load_factor_pct
Floor area	12	m²	Master data → Machines · VMC-850 · floor_area_m2
Machine life	10	years	Master data → Machines · VMC-850 · life_years
Residual value	10	%	Master data → Machines · VMC-850 · residual_pct
Maintenance per year	4	% of price	Master data → Machines · VMC-850 · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · CNC milling	1	persons	Master data → Mould & die database → Tool processes · CNC milling · EXAMPLE
Monthly salary / CTC	30,000	INR/month	Rate library #22 · labour_salary / skilled_machinist · IN · 2026-10 · approved · EXAMPLE
Employer extra cost	0	%	Rate library #18 · labour / employer_extra_cost_pct · IN · 2026-10 · approved · EXAMPLE
Supervision & labour overhead	15	%	Rate library #19 · labour / supervision_pct · IN · 2026-10 · approved · EXAMPLE
Productive time	85	%	Rate library #20 · labour / productive_time_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.

Input	Value	Unit	Comes from
Grinding set-up per die	1	h	Master data → Mould & die database → Process time parameters · sheet_metal / grind_setup_h · EXAMPLE
Surface grinding area per hour	300	cm²/h	Master data → Mould & die database → Process time parameters · sheet_metal / grind_cm2_per_h · EXAMPLE
Machine price	1,800,000	INR	Rate library #37 · machine_price / GRIND-SURF · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	80	INR/h	Rate library #38 · consumables / GRIND-SURF · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	7.5	kW	Master data → Machines · GRIND-SURF · power_kw
Power load factor	60	%	Master data → Machines · GRIND-SURF · load_factor_pct
Floor area	8	m²	Master data → Machines · GRIND-SURF · floor_area_m2
Machine life	12	years	Master data → Machines · GRIND-SURF · life_years
Residual value	5	%	Master data → Machines · GRIND-SURF · residual_pct
Maintenance per year	3	% of price	Master data → Machines · GRIND-SURF · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Grinding	1	persons	Master data → Mould & die database → Tool processes · Grinding · EXAMPLE
Smallest die plate height	25	mm	Master data → Mould & die database → Sizing rules · sheet_metal / die_plate_min_mm · EXAMPLE
Die plate height as a multiple of the sheet thickness	12	× t	Master data → Mould & die database → Sizing rules · sheet_metal / die_plate_t_factor · EXAMPLE
Blank length	326.7	mm	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
blank operations in this die	0		Part register · Die 3: Bending (V): operations bend (suggested die split)
trim operations in this die	0		Part register · Die 3: Bending (V): operations bend (suggested die split)
Punch length	60	mm	Master data → Mould & die database → Sizing rules · sheet_metal / punch_len_mm · EXAMPLE

Input	Value	Unit	Comes from
Rate · Heat treatment (per kg)	120	INR/kg	Rate library #613 · process_unit / Heat treatment (per kg) · IN · 2026-10 · approved · EXAMPLE
Minimum charge per job · Heat treatment (per kg)	1,500	INR/job	Rate library #614 · process_min_charge / Heat treatment (per kg) · IN · 2026-10 · approved · EXAMPLE
Fitting per cutting punch	0.5	h	Master data → Mould & die database → Process time parameters · sheet_metal / assy_h_per_punch · EXAMPLE
Fitting & assembly of one die	16	h	Master data → Mould & die database → Process time parameters · sheet_metal / assy_h_base · EXAMPLE
pierce operations in this die	0		Part register · Die 3: Bending (V): operations bend (suggested die split)
Pierced holes	6	pcs	3D file example_bracket_revB.step · rev B · Sheet-metal tools (stage 9) from the 3D model
Machine price	3,500,000	INR	Rate library #43 · machine_price / SPOTTING · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	30	INR/h	Rate library #44 · consumables / SPOTTING · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	7.5	kW	Master data → Machines · SPOTTING · power_kw
Power load factor	30	%	Master data → Machines · SPOTTING · load_factor_pct
Floor area	15	m²	Master data → Machines · SPOTTING · floor_area_m2
Machine life	15	years	Master data → Machines · SPOTTING · life_years
Residual value	10	%	Master data → Machines · SPOTTING · residual_pct
Maintenance per year	2	% of price	Master data → Machines · SPOTTING · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Fitting & assembly	2	persons	Master data → Mould & die database → Tool processes · Fitting & assembly · EXAMPLE
Monthly salary / CTC	45,000	INR/month	Rate library #23 · labour_salary / toolmaker · IN · 2026-10 · approved · EXAMPLE
Employer extra cost	0	%	Rate library #18 · labour / employer_extra_cost_pct · IN · 2026-10 · approved · EXAMPLE

Input	Value	Unit	Comes from
Supervision & labour overhead	15	%	Rate library #19 · labour / supervision_pct · IN · 2026-10 · approved · EXAMPLE
Productive time	85	%	Rate library #20 · labour / productive_time_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
Die design per operation / station	6	h	Master data → Mould & die database → Process time parameters · sheet_metal / design_h_per_op · EXAMPLE
Operations in this die	1		Part register · Die 3: Bending (V): blank / part envelope 326.7 × 80.0 mm
Die design per die	12	h	Master data → Mould & die database → Process time parameters · sheet_metal / design_h_base · EXAMPLE
Operators · Design	1	persons	Master data → Mould & die database → Tool processes · Design · EXAMPLE
Monthly salary / CTC	70,000	INR/month	Rate library #24 · labour_salary / designer · IN · 2026-10 · approved · EXAMPLE
Employer extra cost	0	%	Rate library #18 · labour / employer_extra_cost_pct · IN · 2026-10 · approved · EXAMPLE
Supervision & labour overhead	15	%	Rate library #19 · labour / supervision_pct · IN · 2026-10 · approved · EXAMPLE
Productive time	85	%	Rate library #20 · labour / productive_time_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
Cavities per tool	1	pcs	Your input · Tool features
Net weight of one part	199.8304	g	Part register · Net weight = CAD volume × material density (Materials)
Warranty / spares included (1 = yes)	1		Part register · Default: included
Tools over the programme	1	tools	Part register · Stage C for 200,000 parts/yr (Σ variant annual volume × quantity (STD 200,000 × 1)), 5 programme years: max(tools for capacity 1, tools for tool life 1)
Projected area of one part	0	cm²	Your input · Estimate page
Press safety factor	1.25	×	Master data → Parameters · sheet_metal / press_safety_factor
Force of this die (Σ its operations)	0.5154	t	Part register · Die 3: Bending (V): blank / part envelope 326.7 × 80.0 mm
Machine price	3,000,000	INR	Rate library #63 · machine_price / PRESS-100 · IN · 2026-10 · approved · EXAMPLE
Consumables & tooling per hour	60	INR/h	Rate library #64 · consumables / PRESS-100 · IN · 2026-10 · approved · EXAMPLE
Electricity rate (industrial)	8.5	INR/kWh	Rate library #9 · power / electricity_industrial · IN · 2026-10 · approved · EXAMPLE

Input	Value	Unit	Comes from
Floor space rent	350	INR/m²/month	Rate library #10 · space / rent_industrial · IN · 2026-10 · approved · EXAMPLE
Interest rate	10	%	Rate library #11 · finance / interest_pct · IN · 2026-10 · approved · EXAMPLE
Overhead	15	%	Rate library #12 · overhead / factory_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Working days per year	300	days/yr	Rate library #14 · hours / working_days · IN · 2026-10 · approved · EXAMPLE
Shifts per day	2	shifts	Rate library #15 · hours / shifts_per_day · IN · 2026-10 · approved · EXAMPLE
Hours per shift	8	h	Rate library #16 · hours / hours_per_shift · IN · 2026-10 · approved · EXAMPLE
Machine utilisation	80	%	Rate library #17 · hours / machine_utilisation_pct · IN · 2026-10 · approved · EXAMPLE
Connected power	11	kW	Master data → Machines · PRESS-100 · power_kw
Power load factor	60	%	Master data → Machines · PRESS-100 · load_factor_pct
Floor area	8	m²	Master data → Machines · PRESS-100 · floor_area_m2
Machine life	15	years	Master data → Machines · PRESS-100 · life_years
Residual value	10	%	Master data → Machines · PRESS-100 · residual_pct
Maintenance per year	4	% of price	Master data → Machines · PRESS-100 · maintenance_pct
INR per INR	1	INR/INR	INR is the base currency.
Operators · Tryout	1	persons	Master data → Mould & die database → Tool processes · Tryout · EXAMPLE
Number of trials	2	trials	Master data → Parameters · sheet_metal / trials
Machine hours per trial	3	h	Master data → Parameters · sheet_metal / trial_hours_each
Shots per trial	200	shots	Master data → Parameters · sheet_metal / trial_shots_each
Part material rate (trial shots)	72	INR/kg	Rate library #585 · material / Galvanised steel GI · IN · 2026-10 · approved · EXAMPLE
Toolmaker overhead	12	%	Rate library #1341 · tool_commercial / tool_overhead_pct · IN · 2026-10 · approved · EXAMPLE
Toolmaker profit	10	%	Rate library #1342 · tool_commercial / tool_profit_pct · IN · 2026-10 · approved · EXAMPLE
Packing	1	%	Rate library #1343 · tool_commercial / tool_packing_pct · IN · 2026-10 · approved · EXAMPLE
Warranty / spares	3	%	Rate library #1344 · tool_commercial / tool_warranty_pct · IN · 2026-10 · approved · EXAMPLE
Transport per kg	8	INR/kg	Rate library #635 · process_unit / Transport (per kg) · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.
INR per INR	1	INR/INR	INR is the base currency.

Rate versions used (the report can be reproduced with these)

Id	Rate	Country	Month	Value	Unit	Status	Approved by	
9	power / electricity_industrial	IN	2026-10	8.5	INR/kWh	approved	Example seed	EXAMPLE
10	space / rent_industrial	IN	2026-10	350	INR/m²/month	approved	Example seed	EXAMPLE
11	finance / interest_pct	IN	2026-10	10	%	approved	Example seed	EXAMPLE
12	overhead / factory_overhead_pct	IN	2026-10	15	%	approved	Example seed	EXAMPLE
14	hours / working_days	IN	2026-10	300	days/yr	approved	Example seed	EXAMPLE
15	hours / shifts_per_day	IN	2026-10	2	shifts	approved	Example seed	EXAMPLE
16	hours / hours_per_shift	IN	2026-10	8	h	approved	Example seed	EXAMPLE
17	hours / machine_utilisation_pct	IN	2026-10	80	%	approved	Example seed	EXAMPLE
18	labour / employer_extra_cost_pct	IN	2026-10	0	%	approved	Example seed	EXAMPLE
19	labour / supervision_pct	IN	2026-10	15	%	approved	Example seed	EXAMPLE
20	labour / productive_time_pct	IN	2026-10	85	%	approved	Example seed	EXAMPLE
22	labour_salary / skilled_machinist	IN	2026-10	30,000	INR/month	approved	Example seed	EXAMPLE
23	labour_salary / toolmaker	IN	2026-10	45,000	INR/month	approved	Example seed	EXAMPLE
24	labour_salary / designer	IN	2026-10	70,000	INR/month	approved	Example seed	EXAMPLE
27	machine_price / VMC-850	IN	2026-10	4,500,000	INR	approved	Example seed	EXAMPLE
28	consumables / VMC-850	IN	2026-10	150	INR/h	approved	Example seed	EXAMPLE
33	machine_price / WEDM	IN	2026-10	6,000,000	INR	approved	Example seed	EXAMPLE
34	consumables / WEDM	IN	2026-10	250	INR/h	approved	Example seed	EXAMPLE
37	machine_price / GRIND-SURF	IN	2026-10	1,800,000	INR	approved	Example seed	EXAMPLE
38	consumables / GRIND-SURF	IN	2026-10	80	INR/h	approved	Example seed	EXAMPLE
43	machine_price / SPOTTING	IN	2026-10	3,500,000	INR	approved	Example seed	EXAMPLE
44	consumables / SPOTTING	IN	2026-10	30	INR/h	approved	Example seed	EXAMPLE
63	machine_price / PRESS-100	IN	2026-10	3,000,000	INR	approved	Example seed	EXAMPLE
64	consumables / PRESS-100	IN	2026-10	60	INR/h	approved	Example seed	EXAMPLE
585	material / Galvanised steel GI	IN	2026-10	72	INR/kg	approved	Example seed	EXAMPLE
598	tool_material / C45 / S50C	IN	2026-10	110	INR/kg	approved	Example seed	EXAMPLE
599	tool_material / D2 (1.2379)	IN	2026-10	500	INR/kg	approved	Example seed	EXAMPLE

Id	Rate	Country	Month	Value	Unit	Status	Approved by	
602	tool_material / Mould base steel (4140 / 1.7225)	IN	2026-10	140	INR/kg	approved	Example seed	EXAMPLE
613	process_unit / Heat treatment (per kg)	IN	2026-10	120	INR/kg	approved	Example seed	EXAMPLE
614	process_min_charge / Heat treatment (per kg)	IN	2026-10	1,500	INR/job	approved	Example seed	EXAMPLE
635	process_unit / Transport (per kg)	IN	2026-10	8	INR/kg	approved	Example seed	EXAMPLE
1341	tool_commercial / tool_overhead_pct	IN	2026-10	12	%	approved	Example seed	EXAMPLE
1342	tool_commercial / tool_profit_pct	IN	2026-10	10	%	approved	Example seed	EXAMPLE
1343	tool_commercial / tool_packing_pct	IN	2026-10	1	%	approved	Example seed	EXAMPLE
1344	tool_commercial / tool_warranty_pct	IN	2026-10	3	%	approved	Example seed	EXAMPLE

Master-data rows still marked EXAMPLE

BOM elements · Backing plate (hardened), BOM elements · Bending die (V-block), BOM elements · Bending punch, BOM elements · Blanking / trimming punch, BOM elements · CNC machining - plates & forming steels, BOM elements · Die design, BOM elements · Die plate (cutting), BOM elements · Die springs / urethane, BOM elements · Fitting & assembly, BOM elements · Heat treatment (die steels), BOM elements · Piercing punch (standard), BOM elements · Pressure pad, BOM elements · Punch holder plate, BOM elements · Screws, dowels & bolts (set), BOM elements · Stock guides & stopper, BOM elements · Stripper plate, BOM elements · Surface grinding, BOM elements · Wire cut - die profile, BOM elements · Wire cut - punches, Machines · GRIND-SURF, Machines · PRESS-100, Machines · SPOTTING, Machines · VMC-850, Machines · WEDM, Mould bases · 4-pillar die set 5040, Process time parameters · sheet_metal / assy_h_base, Process time parameters · sheet_metal / assy_h_per_punch, Process time parameters · sheet_metal / cnc_plate_cm2_per_h, Process time parameters · sheet_metal / cnc_setup_h, Process time parameters · sheet_metal / design_h_base, Process time parameters · sheet_metal / design_h_per_op, Process time parameters · sheet_metal / grind_cm2_per_h, Process time parameters · sheet_metal / grind_setup_h, Process time parameters · sheet_metal / wirecut_mm2_per_h, Process time parameters · sheet_metal / wirecut_setup_h, Sizing rules · sheet_metal / backing_h_mm, Sizing rules · sheet_metal / die_plate_min_mm, Sizing rules · sheet_metal / die_plate_t_factor, Sizing rules · sheet_metal / die_wall_mm, Sizing rules · sheet_metal / punch_len_mm, Sizing rules · sheet_metal / punch_plate_h_mm, Sizing rules · sheet_metal / stripper_h_mm, Tool materials · C45 / S50C, Tool materials · D2 (1.2379), Tool materials · Mould base steel (4140 / 1.7225), Tool processes · CNC milling, Tool processes · Design, Tool processes · Fitting & assembly, Tool processes · Grinding, Tool processes · Tryout, Tool processes · Wire cut

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